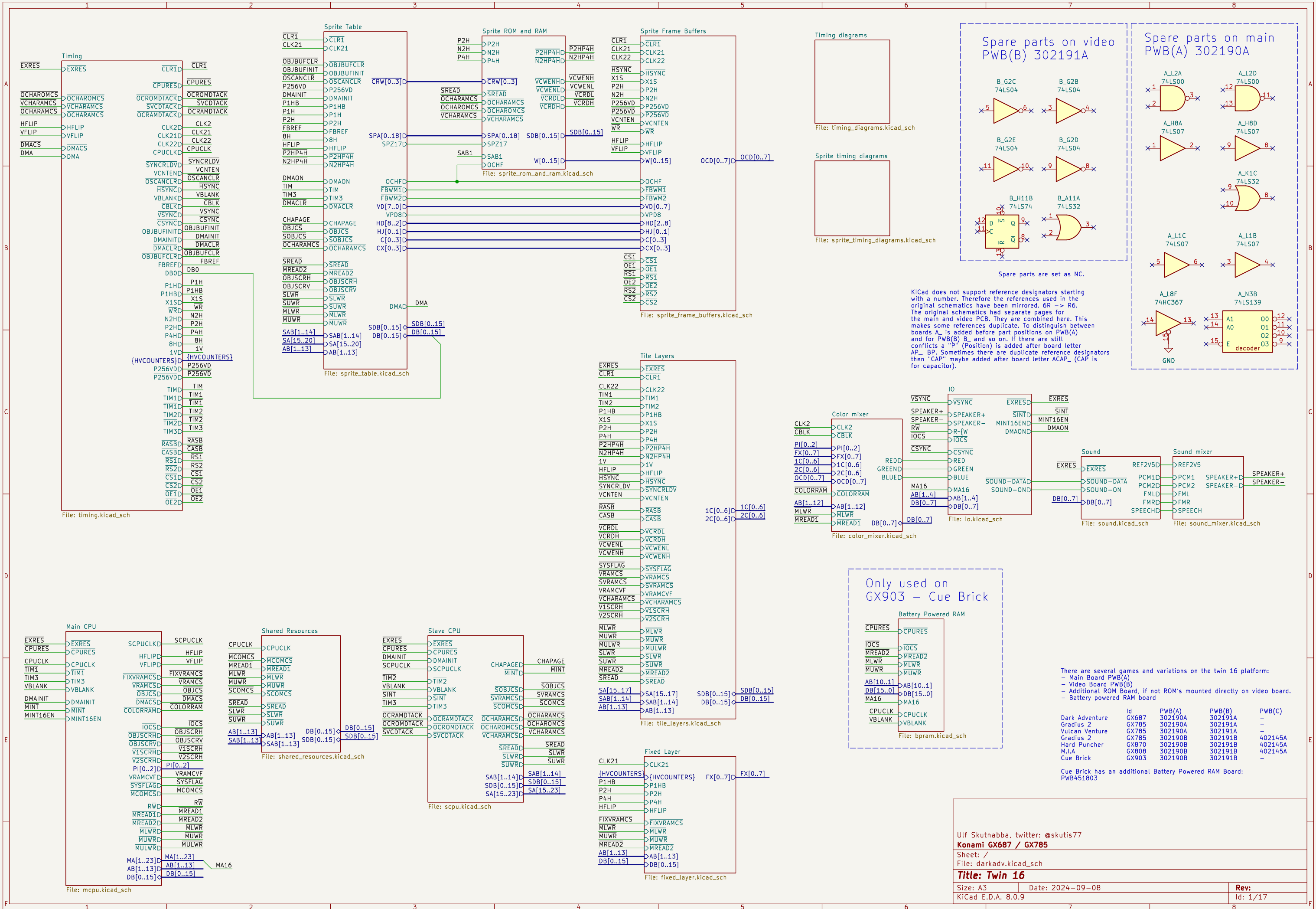
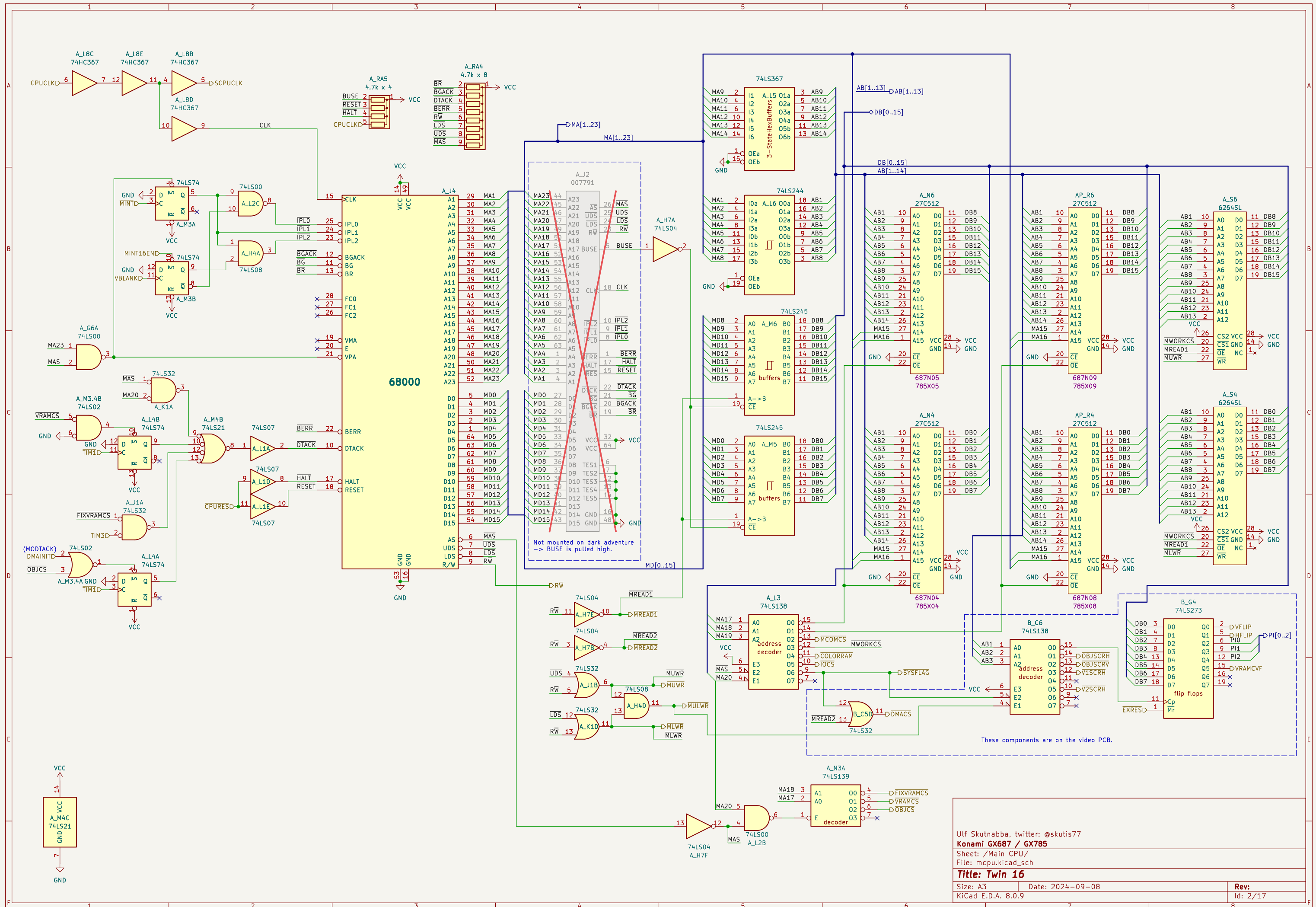
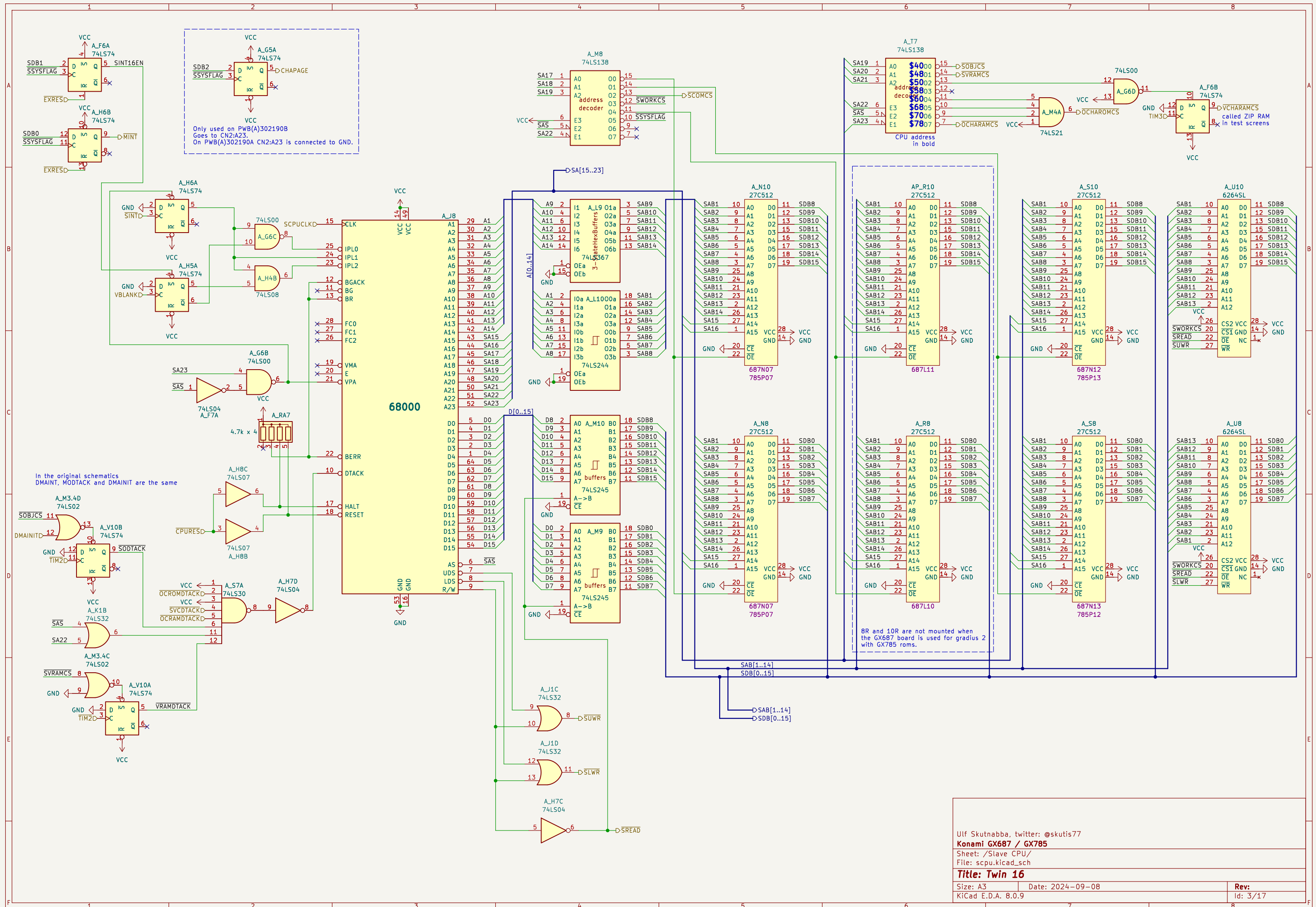
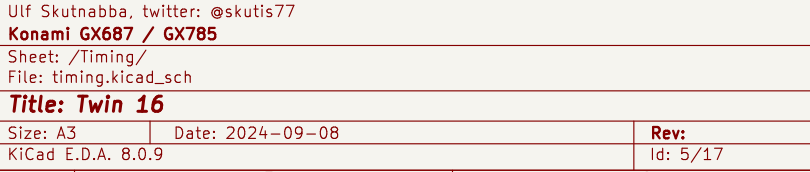
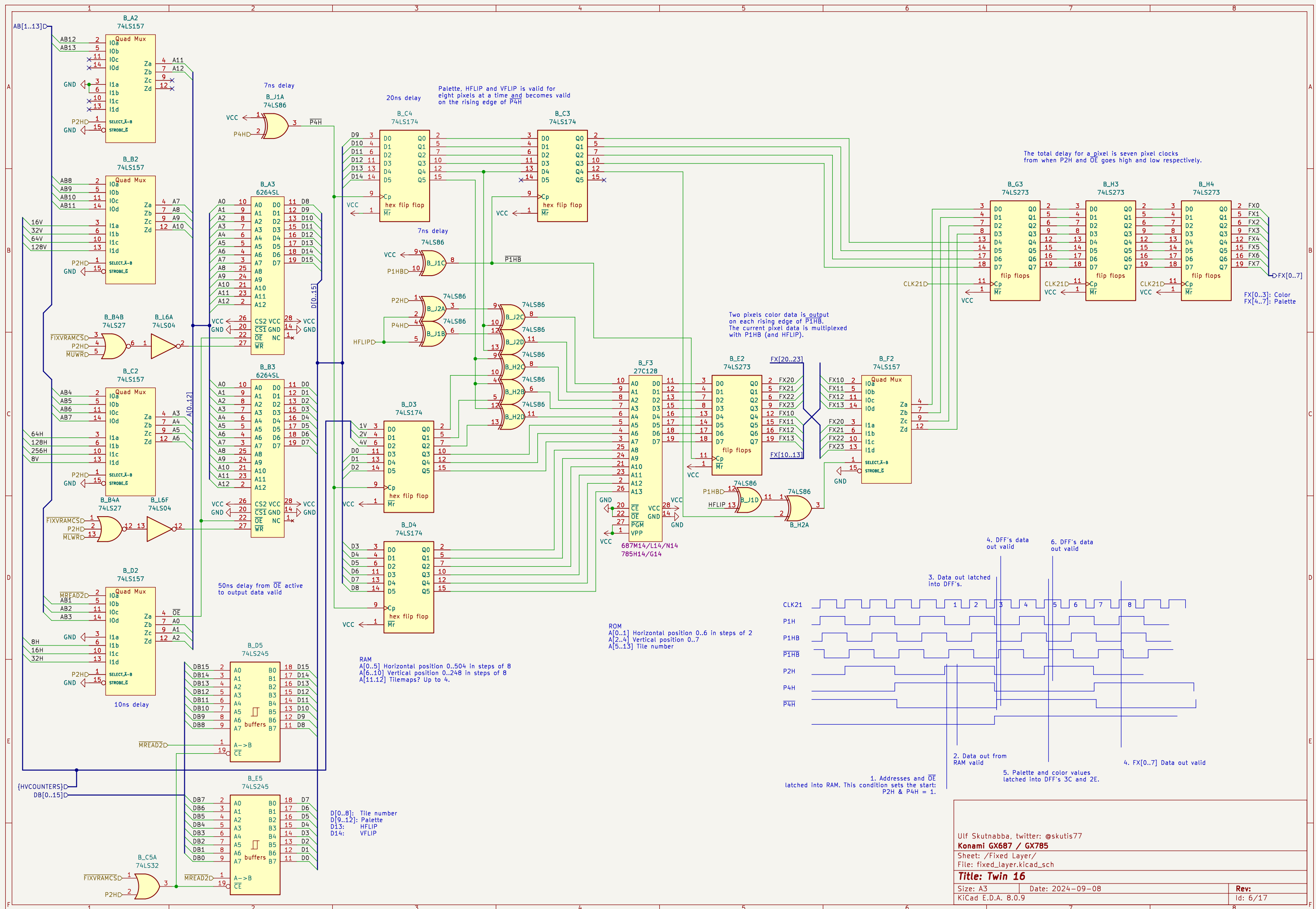


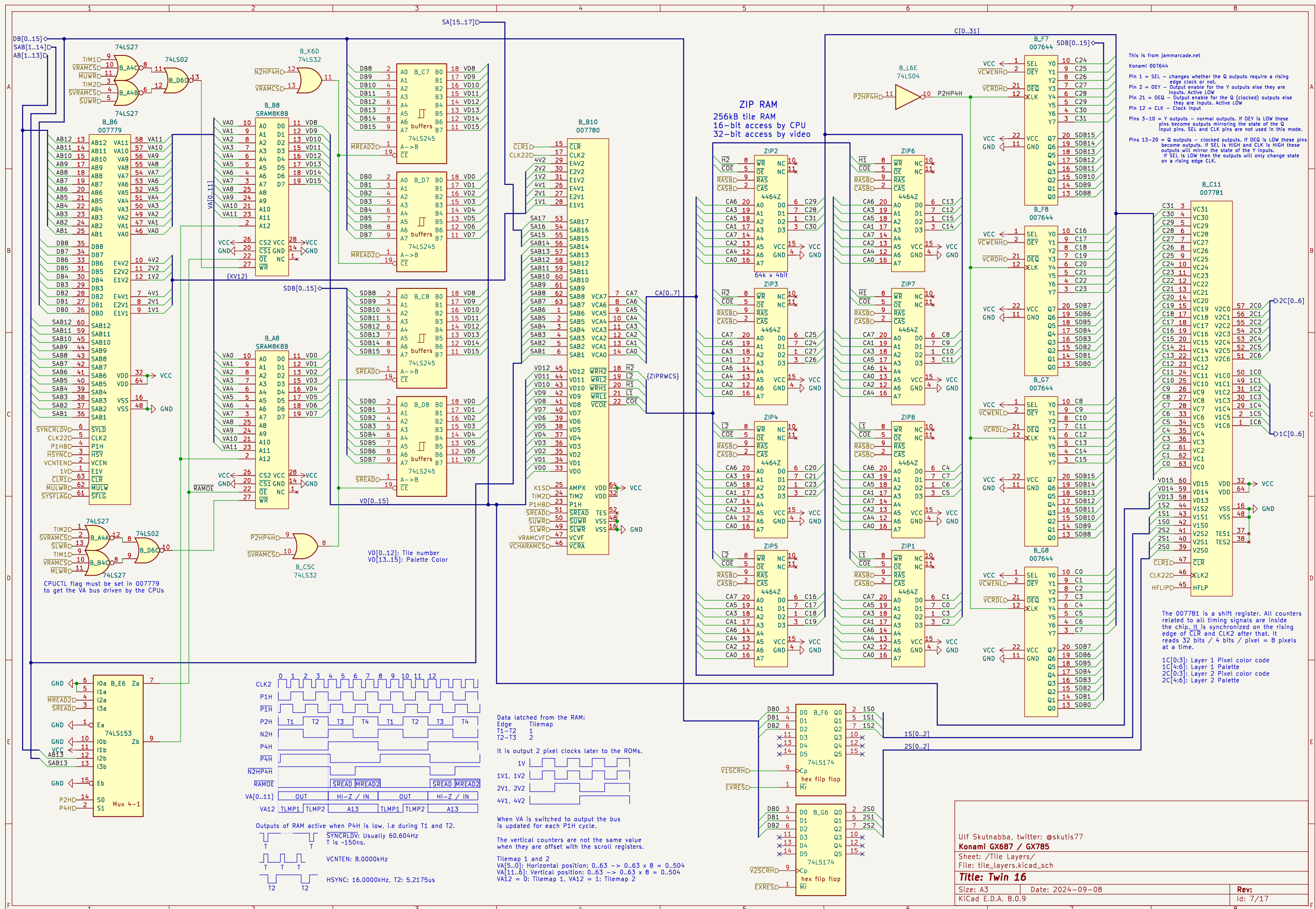


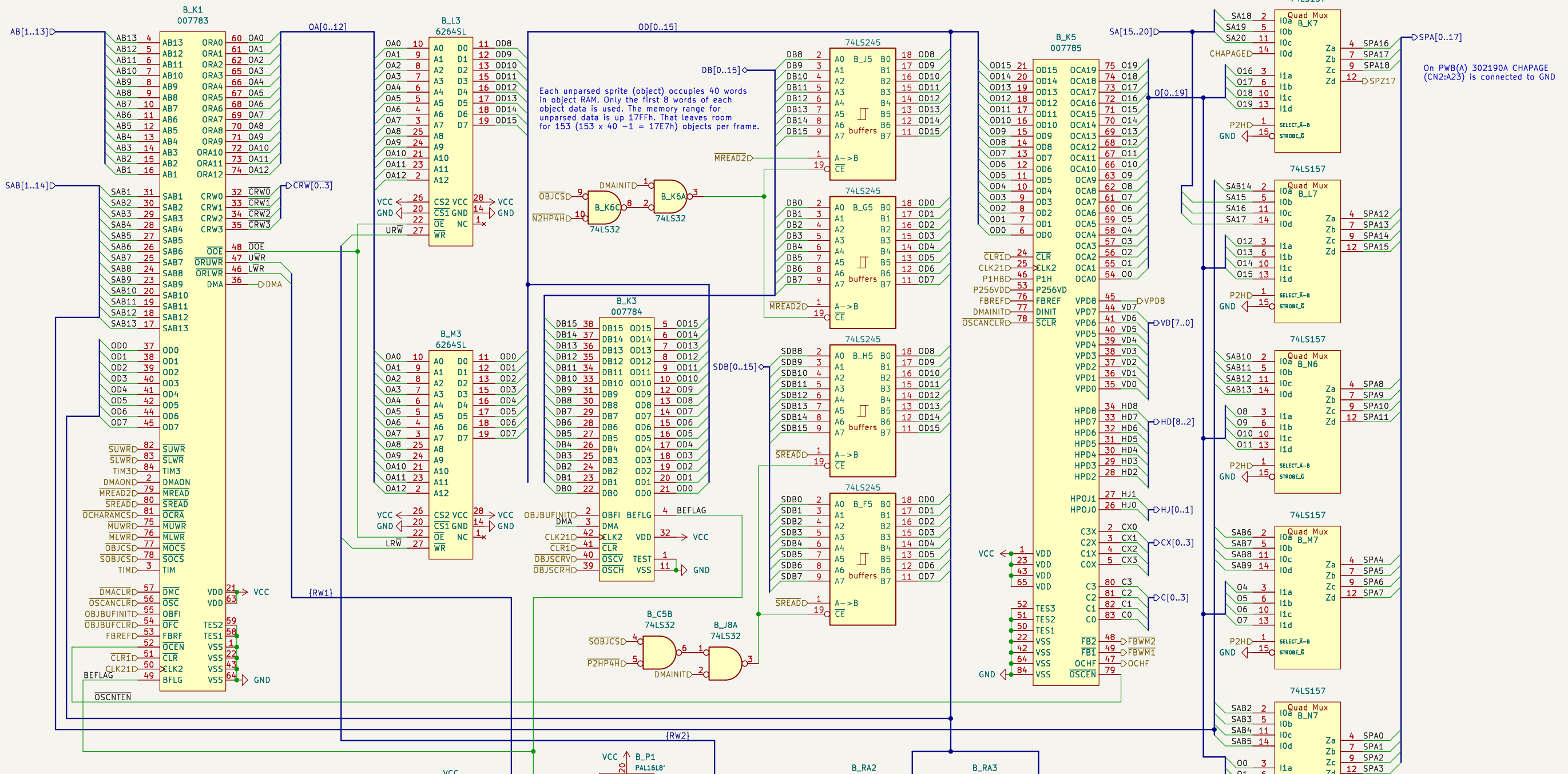












007784 Sprite parser

- subtracts the scroll value for the sprites from current coordinates for the sprites.
- When DMA goes high, the 007784 will read 8 words, if bit OD15 of the first word is high then BEFLAG goes high and stays high until the sprite RAM has processed the first four words and possibly updated the four following words at the high end of the RAM. It will compress the eight words into four words and also subtract the x- and y-scroll values from the x and y coordinates. It will write the four words at the high end of the sprite RAM.
- Addresses at the low end goes from 0x0000..0x0007 and the following cycle at 0x0028..0x002F..

Scroll Register

Register	Sprite horizontal scroll	Sprite vertical scroll
OBJSCRH		
OBJSCRH		

Data is latched in the scroll registers when the signals OBJSCRH and OBJSCRH go low.

007793

CLK21' = CLK21

LRW = -(DMAONS & OBJBUFINIT & CLK21' | BEFLAG & DMA & P1H & DMAONS & CLK21' & (-BH & HFLIP | 8H & -HFLIP) -LRW & -DMAINIT)

URW = -(DMAONS & OBJBUFINIT & CLK21' | BEFLAG & DMA & P1H & DMAONS & CLK21' & (BH & -HFLIP | -BH & HFLIP) -URW & -DMAINIT)

The sprite memory is controlled in three separate cases:

1. OBJBUFINIT
2. BEFLAG & DMA
3. -DMAINIT

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Sheet: /Sprite Table/
File: sprite_table.kicad_sch

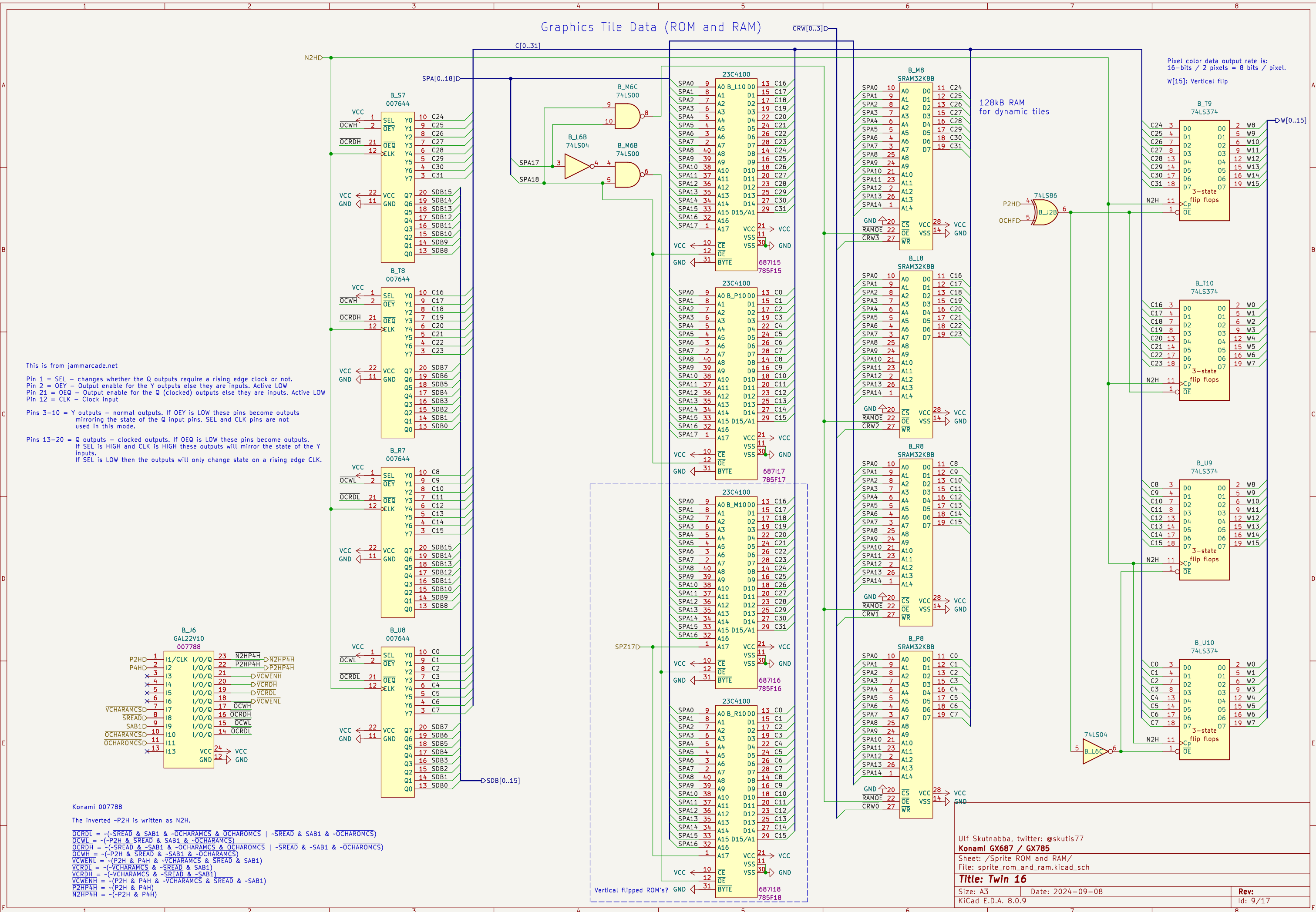
Title: Twin 16

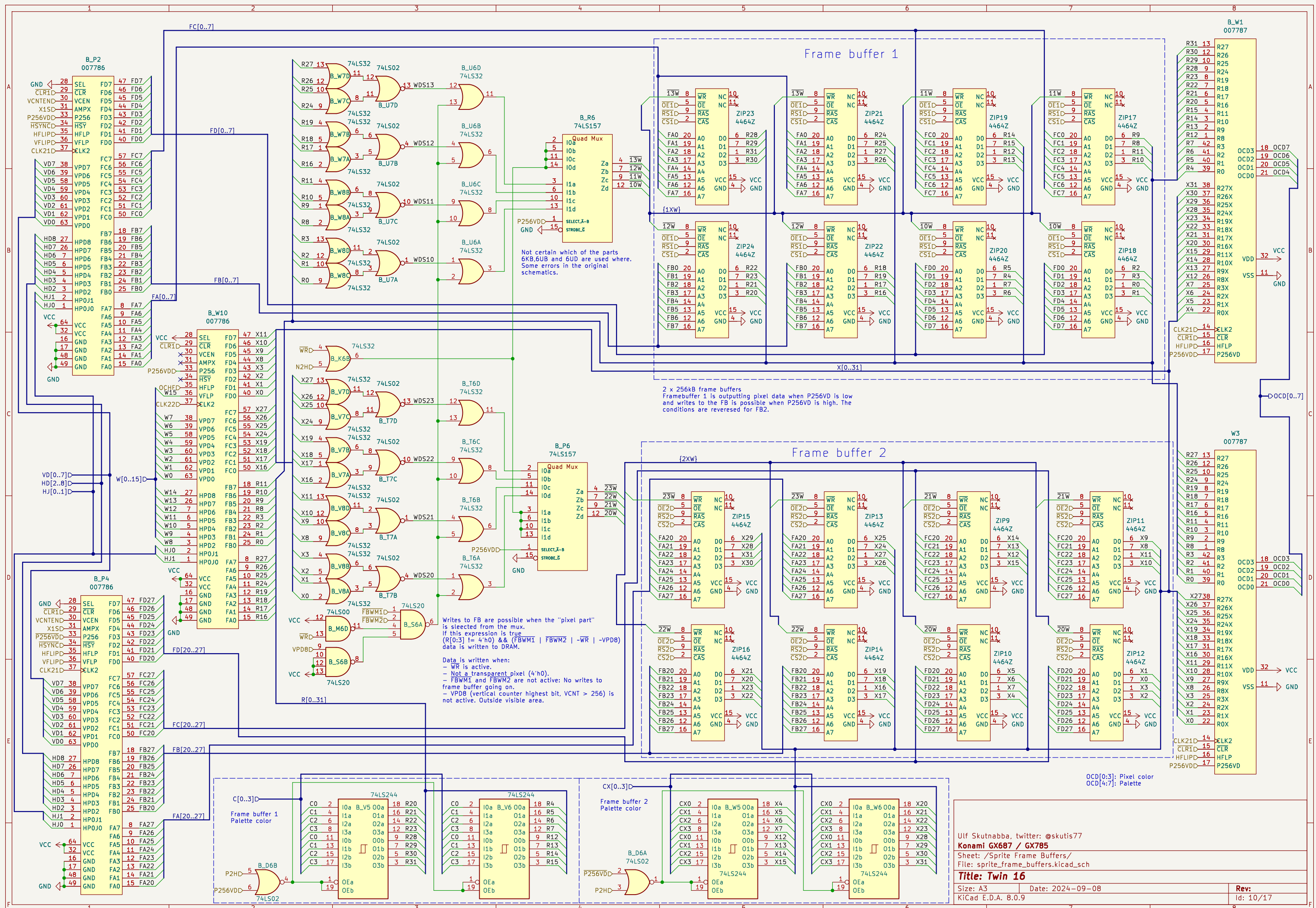
Size: A3 Date: 2024-09-08

KiCad E.D.A. 8.0.9

Rev:
Id: 8/17

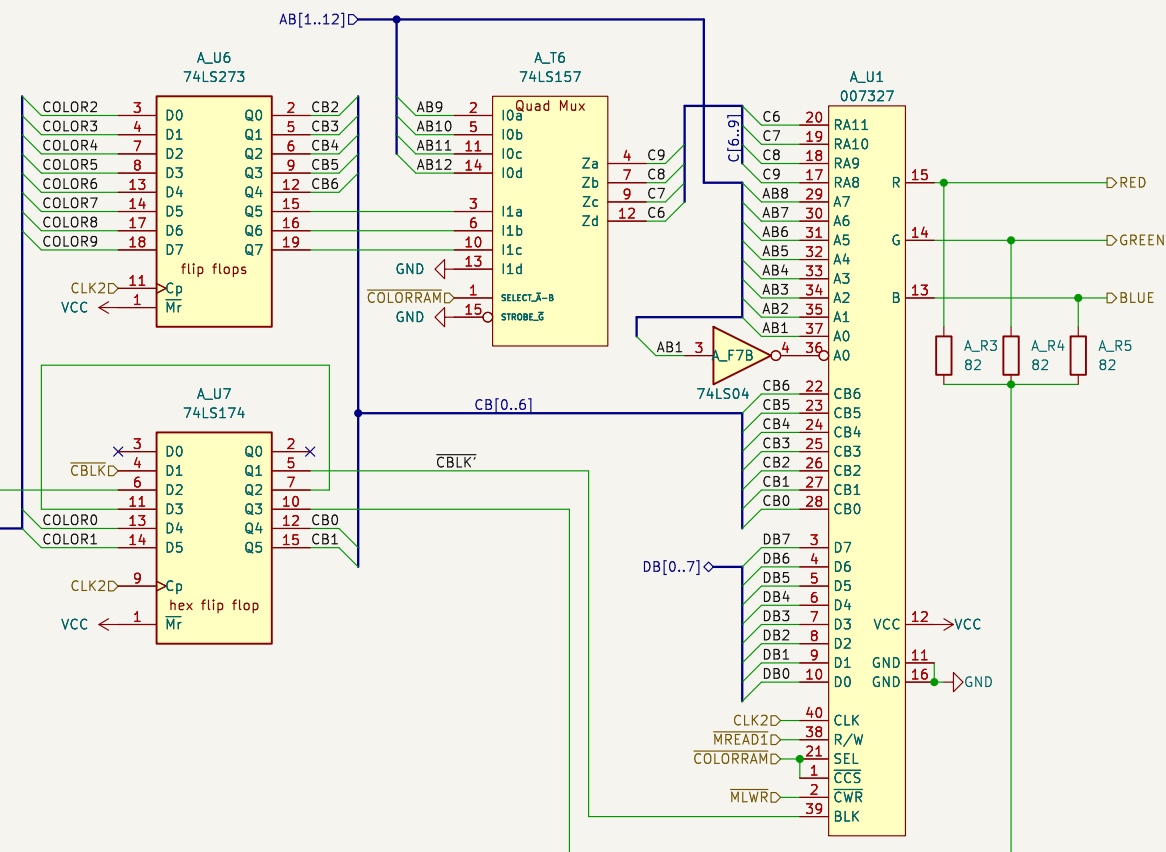
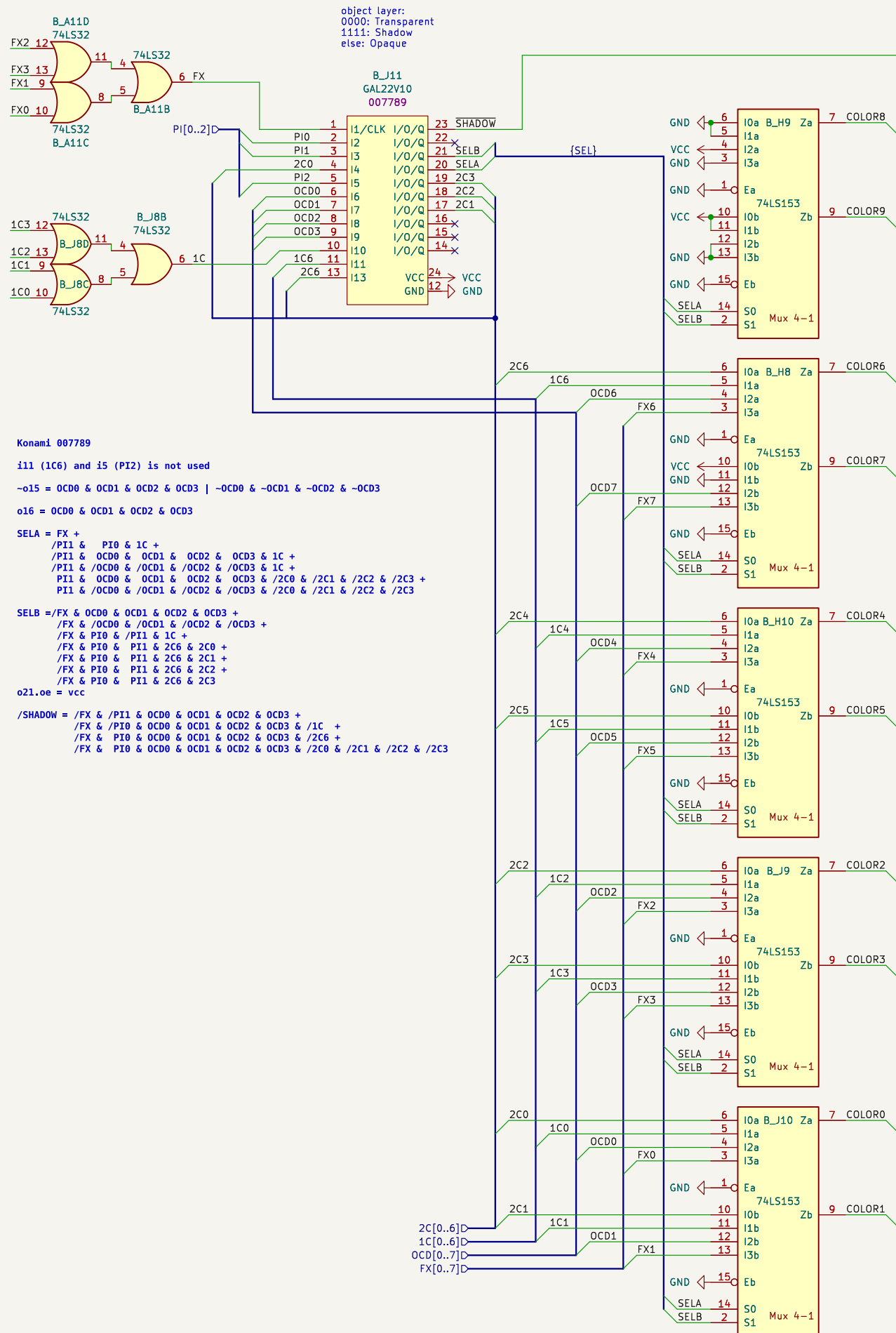
Graphics Tile Data (ROM and RAM)





Components are on Video PCB

Components are on Main PCB



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Sheet: /Color mixer/

File: color_mixer.kicad_sch

Title: Twin 16

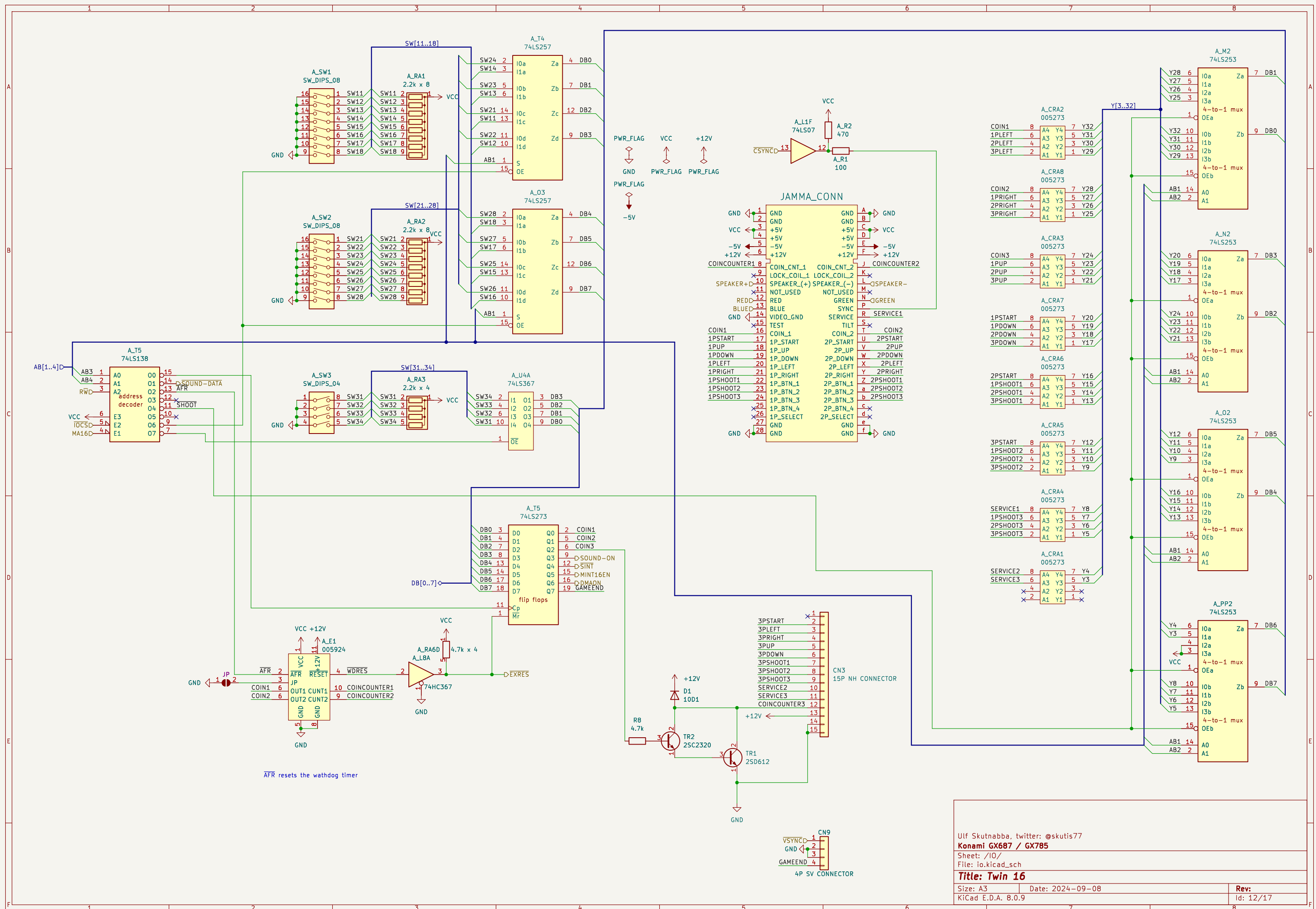
Size: A3

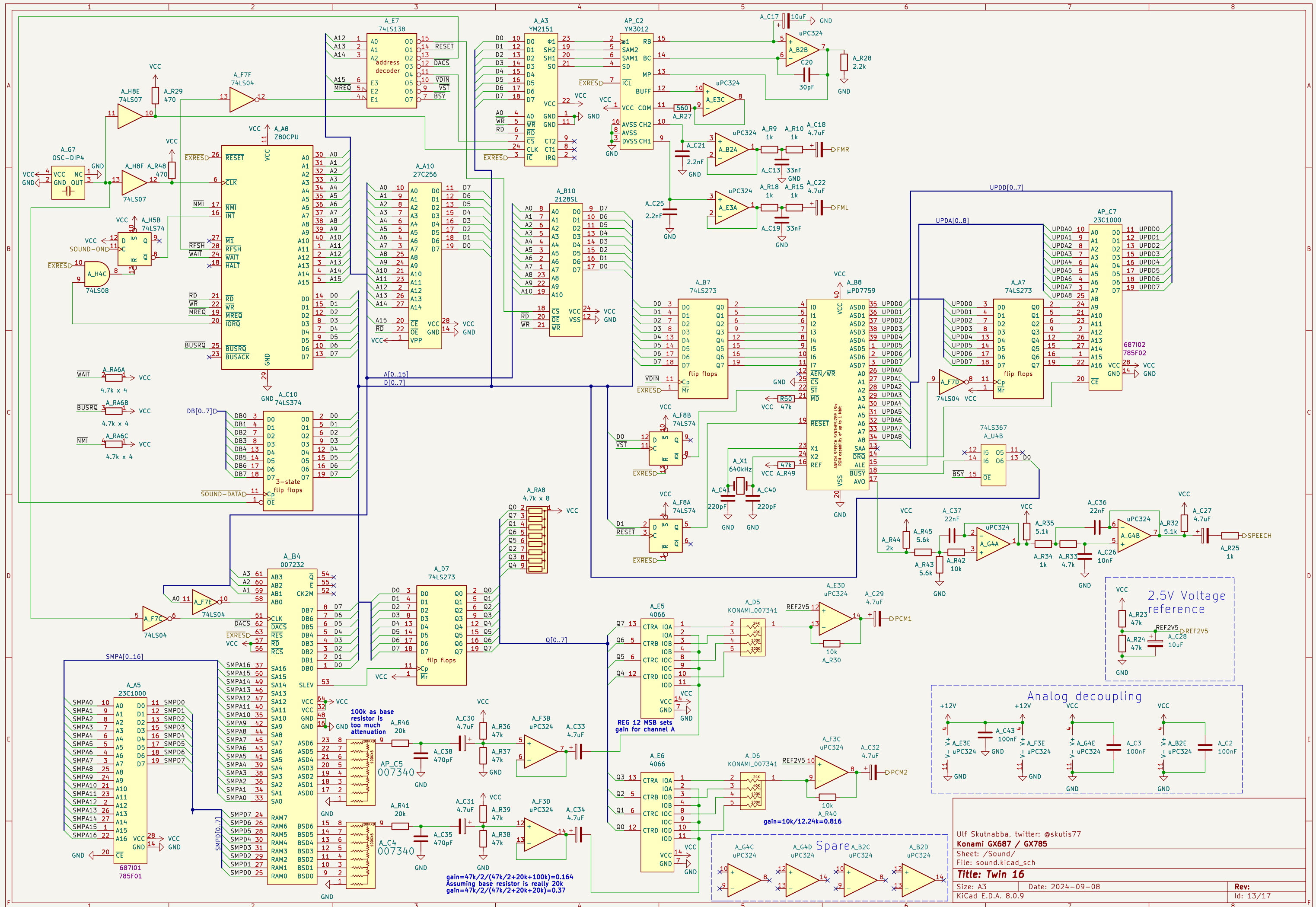
Date: 2024-09-08

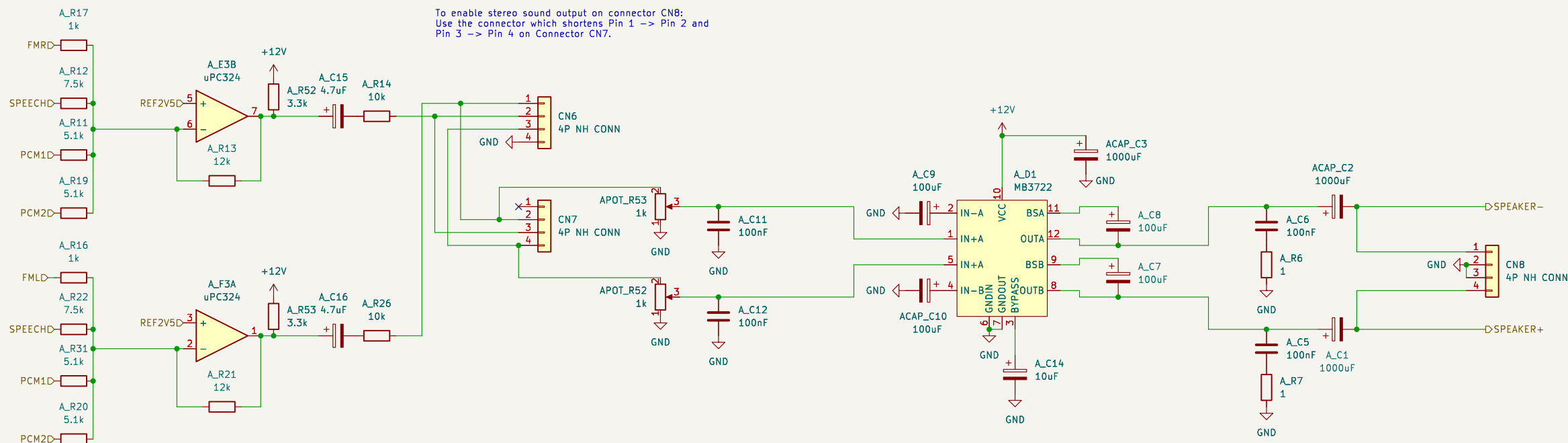
Rev:

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Id: 11/17







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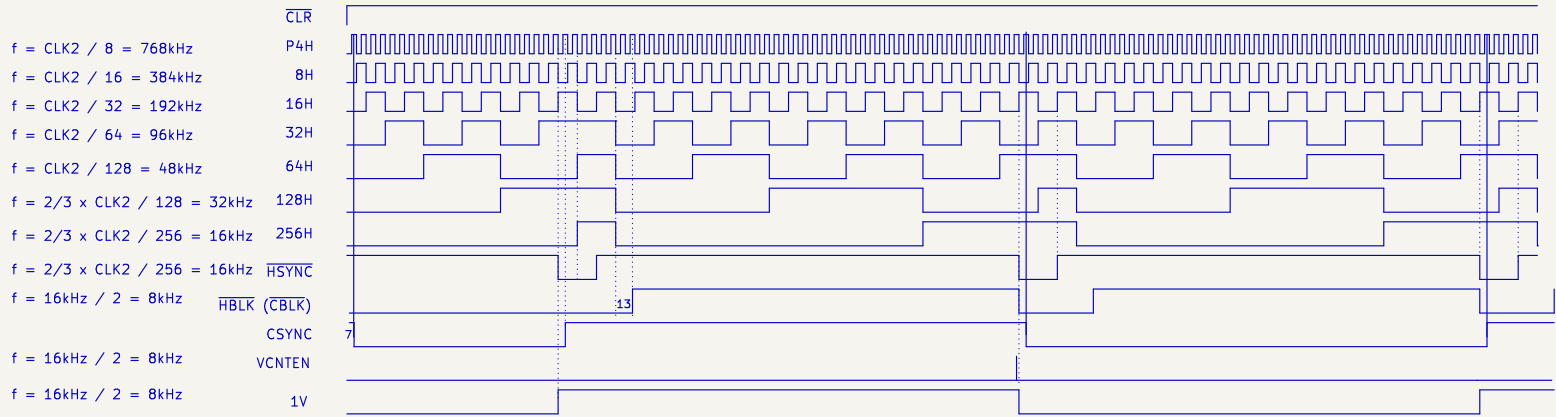
Sheet: /Sound mixer/
File: sound_mixer.kicad_sch

Title: Twin 16

Size: A3 Date: 2024-09-08
KiCad E.D.A. 8.0.9

Rev:
Id: 14/17

Horizontal signals

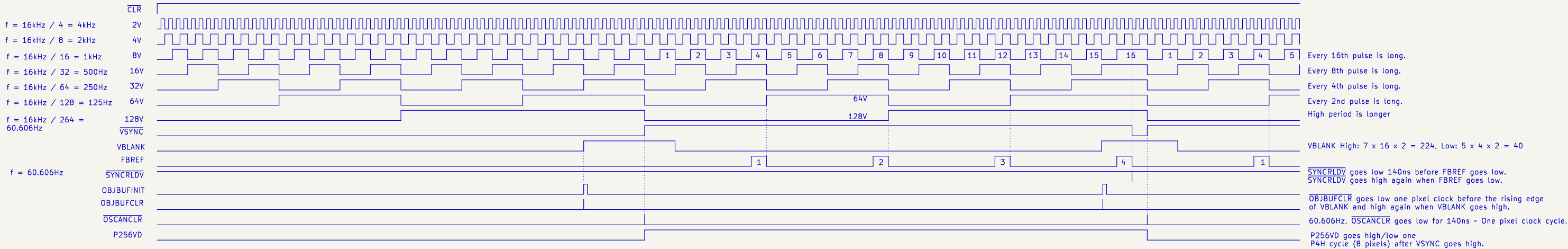
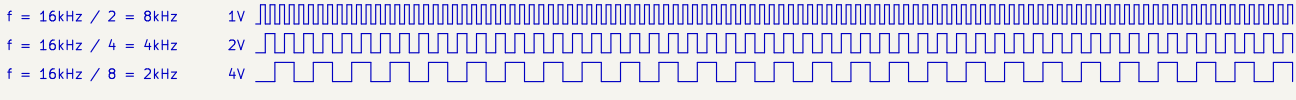


– The first VCNTEN is skipped after reset. It goes low 140ns before HSYNC goes low, and high again when HSYNC goes low. VCNTEN is active right before every second falling edge of HSYNC.

– CPURES goes high, and stays high, on the seventh falling edge of HSYNC.

The first HSYNC is at HCOUNT 175 after reset.

Vertical signals



Every 16th pulse is long.
Every 8th pulse is long.
Every 4th pulse is long.
Every 2nd pulse is long.
High period is longer

VBLANK High: $7 \times 16 \times 2 = 224$, Low: $5 \times 4 \times 2 = 40$

SYNCRLDV goes low 140ns before FBREF goes low. SYNCRLDV goes high again when FBREF goes low.

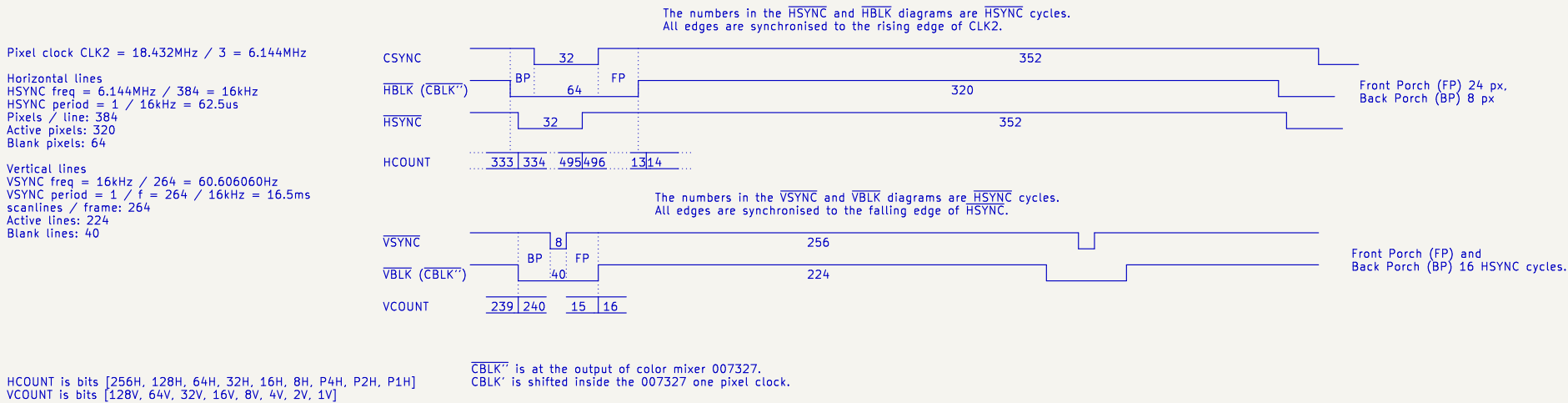
OBJBUFCLR goes low one pixel clock before the rising edge of VBLANK and high again when VBLANK goes high.

60.606Hz, OSCANCLR goes low for 140ns – One pixel clock cycle.

P256VD goes high/low one P4H cycle (8 pixels) after VSYNC goes high.

FBREF is measured at pin 22. At U186:3 FBREF is delayed by 22ns going high and 12ns going low.

Horizontal and vertical synch timing diagrams



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Sheet: /Timing diagrams/
File: timing_diagrams.kicad_sch

Title: Twin 16

Size: A3	Date: 2024-09-08	Rev:
KiCad E.D.A. 8.0.9		Id: 15/17

